

1 Questions

Nick's Question 1.1
In the Wave equation paper you showed that for the Greens function, we have that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}$$

and so we see that

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}(ct, \cdot)(\xi)$$

However in the Fractional SPDE paper the greens function is more complicated and we have that

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \middle| \begin{matrix} (0, 1) \\ (0, 1) \end{matrix} \right. (1 - \beta - \gamma, \beta) \Big) \\ &= t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |\xi|^\alpha \right)^{-s} ds \end{aligned}$$

And so we see that

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |c\xi|^\alpha \right)^{-s} ds$$

However I am unsure how to handle the constant, c , on the inside so we can get a nice scaling property like we do in the Wave Equation paper.

Also, instead of using the Fox-H function we can just say

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

and so

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))}$$

but I still don't see how to handle the c .

Here is how one should proceed:

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta c^\alpha |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} (tc^{\frac{\alpha}{\beta}})^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

$$= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} \mathcal{F}Y_{\alpha,\beta,\gamma,d} \left(c^{\frac{\alpha}{\beta}} t, \cdot \right) (c\xi)$$

Please check if this scaling reduces to heat and wave equations correctly.

Nick's Question 1.2 In the wave equation paper with Xia Chen you said that that fundamental solution to the SPDE

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{1}{2} \Delta u(t, x) = \sqrt{\Theta} u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 \\ u_t(0, \cdot) = 0 \end{cases} \quad (1.1)$$

wspde

and the mild solution is defined to be

$$u(t, x) = 1 + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \quad (1.2)$$

weqs

but for general initial conditions for the wave equation we have the solution to be

$$\begin{aligned} u(t, x) = & \frac{\partial}{\partial t} \int_{\mathbb{R}^d} G(t-s, x-y) u(0, y) dy + \int_{\mathbb{R}^d} G(t-s, x-y) u_t(0, y) dy \\ & + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \end{aligned} \quad (1.3)$$

gweqs

But when the initial conditions are such as in 1.1 then the first two integral in 1.3 are 1 and 0 respectively and so we arrive at 1.2.

In the fractional SPDE

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2] \end{cases} \quad (1.4)$$

fspde

and we define

$$J_0(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) & \beta \in (0, 1] \\ (Z^*(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) & \beta \in (1, 2] \end{cases}$$

and

$$I(t, x) = \iint_{[0,t] \times \mathbb{R}} Y(t-s, x-y) \rho(u(s, y)) W(\delta y) ds$$

and the general solution for 1.4 is

$$u(t, x) = J_0(t, x) + I(t, x) \quad (1.5)$$

feqs

My question is that if we consider the following fractional spde

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases} \quad (1.6)$$

fspde1

then can we calculate $J_0(t, x)$? More specifically is $J_0(t, x) = 1$ in this case? The wave equation paper worked out nicely because in that paper $J_0(t, x) = 1$ and so the solution was able to be written as 1.2. The 1 allows us to easily do the Piccard iteration to get equation (2.3) from the Wave equation paper with Xia Chen.

So if our solution can not be written as

$$u(t, x) = 1 + I(t, x)$$

I am not sure at the moment how we can come up with an expression for the f'_n s in the Weiner Expansion like you did in equation (2.3) in the paper with Xia Chen.

Hi, Nick, here is the answer:

Notice that, by Theorem 4.1 of my paper,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}(t, \cdot)(0) \\ &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil}(0) \\ &= \frac{t^{\lceil \beta \rceil - 1}}{\Gamma(\lceil \beta \rceil)}. \end{aligned}$$

Hence, when $\beta \in (0, 1]$, we see that $(Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) = 1$. When $\beta \in (1, 2]$,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}^*(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}^*(t, \cdot)(0) \\ &= E_{\beta}(0) \\ &= \frac{1}{\Gamma(1)} = 1. \end{aligned}$$

Hence, under the initial conditions given by (1.6), $J_0(t, x) \equiv 1$.

Nick's Question 1.3 Guess for the constant (1.20) from the Wave Equation Paper In the wave equation paper, the constant (1.20) was

$$C = \begin{cases} \frac{4 - \Delta}{3 - \Delta} & \text{under Assumption A} \\ \frac{4 - d}{3 - d} & \text{under Assumption B or C} \end{cases} \quad (1.7) \quad \boxed{\text{const}}$$

I believe in the fractional paper the constant will be

$$\begin{cases} \frac{2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha}{2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha} & \text{under Assumption A} \\ \frac{2(\beta + \gamma - 1) + 2 - \beta d/\alpha}{2(\beta + \gamma - 1) + 1 - \beta d/\alpha} & \text{under Assumption B or C} \end{cases}$$

I think this because in the wave equation paper where we use $(4 - \Delta)$ this gets replaced by $(2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha)$ and thus we can guess that

$$\begin{cases} 4 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha \\ 3 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha \end{cases}$$

Note that when $\alpha = \beta = 2$ and when $\gamma = 0$ then the guess reduces to the wave equation.

Nick's Question 1.4 Lemma 2.3 for the fractional equation

1. In the outline you sent me, you use

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

Where $H^a(\mathbb{R}^d)$ is the fractional Sobelov Space where $H^2(\mathbb{R}^d) = W^{1,2}(\mathbb{R}^d)$. However, I saw on line that $H^a(\mathbb{R}^d) = W^{a,2}(\mathbb{R}^d)$ and so $H^2(\mathbb{R}^d) = W^{2,2}(\mathbb{R}^d)$. Is there a mistake here?

2. Where is $\mathcal{M}$ defined with the sup over $\mathcal{F}_d$ and not $\mathcal{F}_a$? Is this a typo?
3. When defining $\mathcal{E}_a(g, g)$ why does the $(2\pi)^{-d}$ appear here but not in the wave paper?
4. How do we generalize ρ from the wave paper to the fractional equation? For the wave equation we have that

$$\rho = (\mathcal{M}(\gamma, 2))^{(4-\alpha)/2}$$

but why does it generalize to this in the fractional sense

$$\rho = (\mathcal{M}_a(\gamma, 2))^{(2a-\alpha)/a} = 2^{-a/\alpha} (\mathcal{M}_a(\gamma, 1))^{(2a-\alpha)/a}$$

which is what you put in the outline you sent me.

Nick's Question 1.5 : Typos from Xia Chen Paper (Large deviations for Riesz potentials of additive processes)

From what I can tell there seem to be several typos that I can't justify from this paper that seem serious. The following are some definitions and a Theorem from Xia Chen's Paper.

Definitions:

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(s+t)f(t)}{\sqrt{1+|s+t|^\beta} \sqrt{1+|t|^\beta}} dt \right] \varphi_{d-\alpha}(s) ds \quad (1.8)$$

$$\mathcal{E}_\beta(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |s|^\beta |\hat{f}(s)|^2 ds \quad (1.9)$$

$$\mathcal{F}_\beta = \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1, \mathcal{E}_\beta(f, f) < \infty \right\} \quad (1.10)$$

$$\Lambda_\alpha = \sup_{f \in \mathcal{F}_\beta} \left\{ \left(\int_{\mathbb{R}^d} \frac{\prod_{j=1}^p f^2(x_j)}{|x_1 + \dots + x_p|^\alpha} \prod_{j=1}^p dx_j \right)^{1/p} - \mathcal{E}_\beta(f, f) \right\} \quad (1.11)$$

Theorem 1.5

$$\rho = (2\pi)^{-d} (\Lambda_\alpha)^{p-(\alpha/\beta)}$$

This theorem is proved in section 7 of the paper and starts with two lemmas.

Lemma 7.1: Lemma 7.1 claims that Λ_α is finite, however I believe that there is a typo in the proof. In the very first line of the proof, it references equation (A.2) and (A.2) is

proved by referencing (A.8). I believe that equation (A.8) is wrong. (A.8) says that

$$\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\alpha}} \prod_{j=1}^p dx_j = \int_{\mathbb{R}^{d2}} \frac{f_1(x)(f_2 * \dots * f_p)(y)}{|x - y|^{d-\alpha}} dx dy$$

but if we let $p = 2$ then (A.8) says that

$$\int_{\mathbb{R}^{2d}} \frac{f_1(x_1)f_2(x_2)}{|x_1 + x_2|^{d-\alpha}} dx_1 dx_2 = \int_{\mathbb{R}^{d2}} \frac{f_1(x)f_2(y)}{|x - y|^{d-\alpha}} dx dy$$

which is not true. For this issue, we might need to assume that all f_i are symmetric functions.

Lemma 7.2

From first glance I think Lemma 7.2 is okay.

Proof of Theorem 1.5: First some definitions for this theorem.

Definitions:

1.

$$L(f) = \left(\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \right)^{1/p}$$

2.

$$\tilde{L}(f) = \mathcal{E}_\beta(f, f)$$

3.

$$J = \sup_{\|f\|_2^2 + \tilde{f} = 1} L(f) = \sup_{f \in \mathcal{H}} \frac{L(f)}{\|f\|_2^2 + \tilde{L}(f)}$$

There are several places in the proof that I think are typos. Also I do not understand for sure why we have the equality in the supremums in definition (3) above.

1. First is equation (7.21) which says by Parseval's Relation we get

$$\sup_{\langle g, Qg \rangle = 1} \int_{\mathbb{R}^d} [(Qg) * (\widetilde{Qg})]^p(-s) \varphi(s) ds = \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}([(Qg) * (\widetilde{Qg})]^p)(x) \mathcal{F}\varphi(x) dx \quad (1.12)$$

typo1

where $\tilde{p}(x) = p(-x)$ and $g(x) = f(x)\sqrt{1 + |x|^\beta}$ where f is the same function from the definition of ρ . Notice the left hand side of 1.12 has a $(-s)$ as the input of the function. Then when taking the Fourier transform of $[(Qg) * (\widetilde{Qg})]^p(-s)$ we should have the following

$$\mathcal{F}[(Qg) * (\widetilde{Qg})]^p(-x)$$

which follows from the property that if we let $\tilde{p}(x) = p(-x)$ then $\mathcal{F}(\tilde{p})(\xi) = \mathcal{F}(p)(-\xi)$.

Here is what is going on here: First of all, for $\tilde{p}(x) = p(-x)$,

$$\mathcal{F}\tilde{p}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} p(-x) dx = \int_{\mathbb{R}^d} e^{ix\xi} p(x) dx = \overline{\mathcal{F}f(\xi)}.$$

Hence,

$$\int_{\mathbb{R}^d} \tilde{f}(x)g(x)dx = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi) \overline{\mathcal{F}\tilde{f}(\xi)} d\xi = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi) \mathcal{F}f(\xi) d\xi.$$

Nicks Solution: The centered equation (7.20) in the Xia Chen papers says

$$\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t)dt \right]^p \varphi(s)ds = \int_{\mathbb{R}^d} \left[(Qg) * (\widetilde{Qg}) \right]^p (-s) \varphi(s)ds$$

and this equality is true, but we can continue in a different way. Using a substitution $u = s + t$ we see that

$$\begin{aligned} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t)dt \right]^p \varphi(s)ds &= \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} (Qg)(u)(Qg)(u-s)du \right]^p \varphi(s)ds \\ &= \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} (Qg)(u)(\widetilde{Qg})(s-u)du \right]^p \varphi(s)ds \\ &= \int_{\mathbb{R}^d} \left[(Qg) * (\widetilde{Qg}) \right]^p (s) \varphi(s)ds \end{aligned}$$

and now we can apply Parseval's to see that

$$\int_{\mathbb{R}^d} \left[(Qg) * (\widetilde{Qg}) \right]^p (s) \varphi(s)ds = (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left[(Qg) * (\widetilde{Qg}) \right]^p (x) \mathcal{F}\varphi(x)dx$$

and this gives us equation (7.21). What this actually showed was

$$s \mapsto \int_{\mathbb{R}^d} (Qg)(s+t)(Qg)(t)dt$$

is an even function no matter what (Qg) by just applying the substitution $u = s + t$.

2. I'm going to continue, assuming that the previous typo was actually not a typo. The next typo is the unlabeled centered equation right above (7.22) which says

$$\begin{aligned} \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left[(Qg) * (\widetilde{Qg}) \right]^p (x) \mathcal{F}\varphi(x)dx & \quad (1.13) \quad \boxed{\text{typo2}} \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d(p+1)} \int_{\mathbb{R}^d} [|\widehat{Qg}|^2]^{*p}(x) \mathcal{F}\varphi(x)dx \end{aligned}$$

It is stated in the paper that $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$ and $\mathcal{F}(fg) = (2\pi)^{-d}\mathcal{F}(f) * \mathcal{F}(g)$ and f^{*p} is the p -fold convolution. So using these rules we should get that

$$\begin{aligned} \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F} \left(\left[(Qg) * (\widetilde{Qg}) \right]^p \right) (x) \mathcal{F}\varphi(x)dx \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} (2\pi)^{-pd} \mathcal{F} \left(\left[(Qg) * (\widetilde{Qg}) \right] \right)^{*p} (x) \mathcal{F}\varphi(x)dx \\ &= \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d(1+p)} \int_{\mathbb{R}^d} \left[\widehat{Qg} \widehat{\widetilde{Qg}} \right]^{*p} (x) \mathcal{F}\varphi(x)dx \end{aligned}$$

Xia's arguments are correct:

$$\mathcal{F} \left(f * \widetilde{f} \right) (\xi) = \mathcal{F}(f)(\xi) \mathcal{F}(\widetilde{f})(\xi) = \mathcal{F}(f)(\xi) \overline{\mathcal{F}(f)(\xi)} = |\mathcal{F}(f)(\xi)|^2.$$

It seems as if they are assuming that $\widehat{\widehat{Qg}} = \widehat{Qg}$ which is only true if Qg is an even function.

Nicks Solution: Since Qg is a real valued function then we have the property that

$$\widehat{\widehat{Qg}} = \overline{\widehat{Qg}}$$

and therefore we have that

$$\int_{\mathbb{R}^d} \left[\widehat{Qg} \widehat{\widehat{Qg}} \right]^{*p}(x) \mathcal{F}\varphi(x) dx = \int_{\mathbb{R}^d} \left[|\widehat{Qg}|^2 \right]^{*p}(x) \mathcal{F}\varphi(x) dx$$

which verifies equation (7.22).

3. **Details for (7.22)** Again I am going to continue as if the previous was not a typo. Im giving the details for (7.22) just for clarity. (7.22) says that if we define $h = (2\pi)^{-d} \widehat{Qg}$ then

$$\begin{aligned} & \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d(p+1)} \int_{\mathbb{R}^d} [|\widehat{Qg}|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \\ &= \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{-d(p-1)} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \end{aligned}$$

Since $(2\pi)^d h = \widehat{Qg}$ then this implies that $\hat{h} = \widetilde{Qg} = Q\tilde{g}$ since Q is even and because when considering $\hat{f}(x) = \int \exp(ixs) f(s) ds$ as the definition of the Fourier transform then we have the property that $\hat{\hat{f}} = (2\pi)^d \tilde{f}$. Also we have that

$$\langle g, Qg \rangle = \langle \tilde{g}, \widetilde{Qg} \rangle = \langle \hat{h} Q^{-1}, \hat{h} \rangle = (2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h)$$

since for this definition of the Fourier transform we have $\langle f, g \rangle = (2\pi)^{-d} \langle \hat{f}, \hat{g} \rangle$. Putting this together gives us (7.22).

Here are the arguments: we may assume g to be even functions, indeed the supremum is achieved for even functions g . Now

$$\begin{aligned} \langle g, Qg \rangle &= \langle \tilde{g}, Qg \rangle \\ &= \langle \hat{\tilde{h}}, Q^{-1} \hat{h} \rangle \\ &= \langle \hat{\tilde{h}}, (1 + \psi) \hat{h} \rangle \\ &= \|\hat{f}\|_2^2 + \int_{\mathbb{R}^d} (1 + \psi(\xi)) |\hat{h}(\xi)|^2 d\xi \\ &= \|\hat{f}\|_2^2 + \int_{\mathbb{R}^d} (1 + \psi(\xi)) |\hat{h}(\xi)|^2 d\xi \\ &= (2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h). \end{aligned}$$

4. **Equation (7.23)** I am unsure why a change of variable implies that

$$\sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{-d(p-1)} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx$$

$$= \sup_{\|h\|_2^2 + \tilde{L}(h)=1} (2\pi)^{-d} \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j$$

In particular how the sup limits changed.

Possible Justification: The integrals can be shown to be equal through a change of variables.

$$\begin{aligned} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx &= \int_{\mathbb{R}^d} (|h|^2 * \dots * |h|^2)(x) \frac{1}{|x|^\alpha} dx \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^{d(p-1)}} |h|^2(x - y_1) |h|^2(y_1 - y_2) \dots |h|^2(y_{p-2} - y_{p-1}) |h|^2(y_{p-1}) \prod_{j=1}^{p-1} dy_j \frac{1}{|x|^\alpha} dx \end{aligned}$$

then applying the substitution

$$\begin{pmatrix} x - y_1 = u_1 \\ y_1 - y_2 = u_2 \\ \dots \\ y_{p-2} - y_{p-1} = u_{p-1} \\ y_{p-1} = u_p \end{pmatrix}$$

will transform the integral as

$$\int_{\mathbb{R}^d} (|h|^2 * \dots * |h|^2)(x) \frac{1}{|x|^\alpha} dx = \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j$$

As for the limits of the supremum, note that

$$(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1 \implies \left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h) = 1$$

Therefore using the above definitions of J , $L(f)$ and $\tilde{L}(f)$ we see that

$$\begin{aligned} \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h)=1} (2\pi)^{-d(p-1)} \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \\ &= \sup_{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)=1} (2\pi)^{-d(p-1)} L(h)^p \\ &= (2\pi)^{-d(p-1)} \left[\sup_{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)=1} L(h) \right]^p \\ &= (2\pi)^{-d(p-1)} \left[\sup_{h \in \mathcal{H}} \frac{L(h)}{\left\| (2\pi)^{d/2} h \right\|_2^2 + \tilde{L}((2\pi)^{d/2} h)} \right]^p \\ &= (2\pi)^{-d(p-1)} \left[(2\pi)^d \sup_{h \in \mathcal{H}} \frac{L(h)}{\|h\|_2^2 + \tilde{L}(h)} \right]^p \\ &= (2\pi)^{-d} J^p \end{aligned}$$

which is exactly equation (7.23) in the Xia Chen paper.

How to apply the above to correlations functions other than the Reisz Kernel:

Suppose we have a spectral measure, μ , with spectral density φ with related tempered measure Γ with density γ . In other words, for all Schwartz functions ψ

$$\begin{aligned}\int_{\mathbb{R}^d} dx \gamma(x) \psi(x) &= \int_{\mathbb{R}^d} \Gamma(dx) \psi(x) \\ &= \int_{\mathbb{R}^d} \mu(dx) \mathcal{F}\psi(x) \\ &= \int_{\mathbb{R}^d} dx \varphi(x) \mathcal{F}\psi(x)\end{aligned}$$

also suppose that we have the following scaling properties

$$\gamma(cx) = c^{-\alpha} \gamma(x) \quad \text{and} \quad \varphi(cx) = c^{-(d-\alpha)}$$

We can follow the proof of Theorem 1.5 from the Xia Chen paper with minor modifications to show that

$$\rho = (2\pi)^{-d} \Lambda(1)^{2-\alpha/\beta}$$

where

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(s+t)f(t)}{\sqrt{1+|s+t|^\beta} \sqrt{1+|t|^\beta}} dt \right] \varphi(s) ds$$

and

$$\Lambda(\theta) = \theta^{\beta/(\beta-\alpha/2)} \Lambda(1)$$

and

$$\begin{aligned}\Lambda(1) &= \sup_{\|f\|_2=1} \left\{ \left(\int_{\mathbb{R}^{2d}} |f(x_1)|^2 |f(x_2)|^2 \gamma(x_1+x_2) dx_1 dx_2 \right)^{1/2} - \mathcal{E}_\beta(f, f) \right\} \\ &= \sup_{\|f\|_2=1} \left\{ \left(\int_{\mathbb{R}^d} [|f|^2 * |f|^2](x) \gamma(x) dx \right)^{1/2} - \mathcal{E}_\beta(f, f) \right\} \\ &= \sup_{\|f\|_2=1} \left\{ \langle |f|^2 * |f|^2, \gamma \rangle^{1/2} - \mathcal{E}_\beta(f, f) \right\}\end{aligned}$$

If this is correct then this means that equation (1.7) in the wave paper is slightly off.

Nick's Question 1.6 Addressing the (A.2) and (A.8) Typo from Xia Chen's Paper

(A.2) says the following, for $0 < \sigma < d$ and some constant C ,

$$\left| \int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\sigma}} \prod_{j=1}^p dx_j \right| \leq C^p \prod_{j=1}^p \|f_j\|_{pd/((p-1)d+\sigma)}$$

and the proof of this inequality relies on (A.8) which says

$$\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\alpha}} \prod_{j=1}^p dx_j = \int_{\mathbb{R}^{d^2}} \frac{f_1(x)(f_2 * \dots * f_p)(y)}{|x-y|^{d-\alpha}} dx dy$$

and (A.8) is only true for even functions. In the Xia Chen paper, (A.2) is used to prove (A.3) as well however these equations are only used in two spots.

For (A.2), it is used to show that Γ_σ is finite. It seems like this is unnecessary because in section 7 if Chen's paper, it is proven that $\rho = (2\pi)^{-d}(\Gamma(1))^{p-\sigma/\beta}$ and ρ had already been proven to be finite in the beginning of the paper and so then Γ must be finite. So using (A.2) to independently prove that Γ is finite seems unnecessary at the moment but I am unsure.

(A.3), and also (A.2), is used in the proof of Lemma 2.2 which says that

$$\mathbb{E} [\zeta([0, t_1], \dots, [0, t_p])^m] = \int_{\mathbb{R}^{dm}} \left[\sum_{\sigma \in \Sigma_m} \prod_{k=1}^m \frac{1}{\lambda_{\sigma(1)+\dots+\lambda_{\sigma(k)}}} \right]^p \prod_{k=1}^m \varphi(\lambda_k) d\lambda_k.$$

It seems that once we have the expression on the right hand side that the proofs for the upper and lower bound in Theorem 2.3 work fine. So maybe in our paper this wont be an issue since we do not deal with the left hand side of the above centered equality.

Possible fix for the (A.2) Inequality : In the Wave paper, we only care about ρ from Xia Chen's paper with $p = 2$. We can rewrite (A.2) as

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(y)}{|x+y|^{d-\sigma}} dx dy = \int_{\mathbb{R}^{2d}} \frac{f(u)g(-v)}{|u-v|^{d-\sigma}} dx dy = \int_{\mathbb{R}^{2d}} \frac{f(u)\tilde{g}(v)}{|u-v|^{d-\sigma}} dx dy$$

Then we can apply (1.26) from Xia Chen's paper with $r = 2d/(\sigma+d)$ and $s = 2d/(\sigma+d)$ to obtain

$$\int_{\mathbb{R}^{2d}} \frac{f(u)\tilde{g}(v)}{|u-v|^{d-\sigma}} dx dy \leq C \|f\|_{2d/(\sigma+d)} \|\tilde{g}\|_{2d/(\sigma+d)}$$

but

$$\|\tilde{g}\|_{2d/(\sigma+d)} = \|g\|_{2d/(\sigma+d)}$$

and so

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(y)}{|x+y|^{d-\sigma}} dx dy \leq C \|f\|_{2d/(\sigma+d)} \|\tilde{g}\|_{2d/(\sigma+d)} = C \|f\|_{2d/(\sigma+d)} \|g\|_{2d/(\sigma+d)}$$

and this is the inequality we need but only for $p = 2$.

Nick's Question 1.7 1.26 From The Xia Chen Paper

I am unsure why we can say

$$\int_{\mathbb{R}^{2d}} \frac{F(\lambda)G(\gamma)}{|\lambda-\gamma|^{d-\sigma}} d\lambda d\gamma \leq C \|F\|_r \|G\|_s$$

where $s^{-1} + r^{-1} = 1 + \sigma/d$. It says it follows by page 275 of *Donoghue - Distributions and Fourier transforms* but in that text what is stated is that

$$\int_{\mathbb{R}^{2d}} u(\lambda)v(\gamma)T(\lambda-\gamma)d\lambda d\gamma \leq \|T\|_r \|v\|_{r'} \|u\|_1$$

where I am unsure what r and r' satisfy but T is required to be in $L^r(\mathbb{R}^d)$ and in the case in Xia Chen's paper, $T(x) = \varphi(x) = \frac{1}{|x|^{d-\sigma}}$ is not in $L^r\mathbb{R}^d$. I think in the Donoghue text we have $1/r + 1/r' + 1 = 2$ which would follow from Young's Convolution Inequality.

The equation (1.26) can be written as

$$\int_{\mathbb{R}^d} I_\sigma(F)(\gamma)G(\gamma)d\gamma \leq C \|F\|_r \|G\|_s$$

where

$$I_\sigma(F)(\gamma) = \int_{\mathbb{R}^d} \frac{F(x)}{|\gamma - x|^{d-\sigma}} dx$$

and I tried using the theorem (Hardy Littlewood Fractional Integration Theorem) which says that I_σ maps from L^p to L^q if and only if $1/p - 1/q = \sigma/d$ with $p > 1$ but as of now I am still working on it.

Partial Solution: Let q and s be such that $1/q + 1/s = 1$ then Holders Inequality gives us that

$$\begin{aligned} \int_{\mathbb{R}^{2d}} \frac{F(\lambda)G(\gamma)}{|\lambda - \gamma|^{d-\gamma}} d\lambda d\gamma &= \int_{\mathbb{R}^d} I_\sigma F(\lambda) G(\gamma) d\lambda \\ &\leq \|I_\sigma F\|_q \|G\|_s \end{aligned}$$

Now recalling the Hardy Littlewood Fractional Integral Theorem, I_σ is a bounded linear operator from L_r to L_q if and only if $1/q = 1/r - \sigma/d$. Let r be such. Then we have

$$\|I_\sigma F\|_q \leq C \|F\|_r$$

and putting this together we get that

$$\int_{\mathbb{R}^{2d}} \frac{F(\lambda)G(\gamma)}{|\lambda - \gamma|^{d-\gamma}} d\lambda d\gamma \leq C \|F\|_r \|G\|_s$$

whenever

$$\begin{aligned} 1 &= \frac{1}{q} + \frac{1}{s} \\ &= \left(\frac{1}{r} - \frac{\sigma}{d} \right) + \frac{1}{s} \end{aligned}$$

hence

$$\frac{1}{r} + \frac{1}{s} = 1 + \frac{\sigma}{d}$$

which is what is given in the Xia Chen paper.

Your solutions above look very good.

Nick's Question 1.8 (1.26) for the fractional noise. results to

If we want to apply the

$$\varphi(x) = \prod_{j=1}^d \frac{1}{|x|^{1-\sigma_j}}$$

then we need to be able to prove equation (1.26) for this noise which is

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{F(\gamma)G(\lambda)}{\prod_{j=1}^d |\lambda_j - \gamma_j|^{1-\sigma_j}} d\gamma d\lambda \leq C \|F\|_r \|G\|_s$$

In order to figure that how we need to find out properties about the following operator,

$$L_\sigma f(x) = \int_{\mathbb{R}^d} \frac{f(y)}{\prod_{j=1}^d |x_j - y_j|^{1-\sigma_j}} dy.$$

Specifically can we say that L_σ is a bounded linear operator from L_p to L_q for some related pair (p, q) ?

The issue with copying the proof for the Hardy-Littlewood fractional integral theorem in this situation is that polar coordinates do not seem to be so easily applied in this situation.

Weak Young's Inequality:

The inequality (1.26) from the Xia Chen paper seems to only be important to show that ρ is finite. We may be able to use Young's Weak inequality to do this, which is

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x)g(x-y)g(y)dx dy \right| \leq K_{p,q,r,d} \|f\|_p \|g\|_{q,w} \|h\|_r$$

Where $p, q, r > 1$ and $1/p + 1/q + 1/r = 2$. I am following page 107 section 4.3 of Analysis by Lieb and Loss. The weak L^q space is defined to be the set of functions such that

$$\sup_{\alpha > 0} \alpha |\{x : |f(x)| > \alpha\}|^{1/q} < \infty.$$

This quantity is not a norm on the L_w^q space though. But there is a norm on the space that it is equivalent to the above expression (I am unsure what equivalent means in this situation since this statement is not proven in the Lieb and Loss text)

$$\|f\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |f(x)| dx$$

where $1/q + 1/q' = 1$ and A is a measurable set of finite measure.

It is also claimed in the Lieb and Loss text that

$$\|f\|_{d/\lambda,w} = \frac{d}{d-\lambda} \left[|\mathbb{S}^{d-1}|/d \right]^{\lambda/d} \text{ when } f(x) = |x|^{-\lambda}$$

If we use this but with $\lambda = d - \alpha$ and $q = d/\lambda$ then we get that $1/p + 1/r = 1 + \alpha/d$ and also

$$\int_{\mathbb{R}^{2d}} \frac{f(x)g(x)}{|x-y|^{d-\lambda}} dx dy \leq K \frac{d}{\alpha} \left[|\mathbb{S}^{d-1}|/d \right]^{(d-\alpha)/d} \|f\|_p \|g\|_r.$$

This is another way to prove that ρ from the Xia Chen paper is finite and this way avoids using the Sobelov-Hardy-Littlewood Theorem that I used above. Things need some work here because we need $\|f\|_p$ and $\|g\|_r$ to be finite. In the Xia Chen paper you can see that these norms just reduced to L^2 norms because they have those functions F and G and when they reduce to f and g then the norms become L^2 norms. The same will happen in this case as well too.

We can try to apply the weak Young's inequality to $\varphi(x) = \prod |x_i|^{1-\alpha_i}$. This would require us calculate $\|\varphi\|_{q,w}$ to show that it is in the weak L_w^q space.

Is fractional noise in the weak L^p space?

Let $\varphi(x) = \prod_{j=1}^d |x_j|^{\alpha_j-1}$. We would like for $\varphi \in L_w^{d/(d-\alpha)}$ because then we would apply weak young's inequality with $q = d/(d-\alpha)$ and see that

$$\int_{\mathbb{R}^{2d}} F(x)\varphi(x-y)G(y) \leq \|\varphi\|_{d/(d-\alpha),w} \|F\|_p \|G\|_r$$

where $1/p + 1/r = 1 + \alpha/d$. Then we could follow the Xia Chen paper exactly to show ρ is finite.

From the definition of the weak L^p space on wiki, we need to show that for

$$K_\varphi(t) = \left| \left\{ x : \prod_{j=1}^d |x_j|^{\alpha_j-1} > t \right\} \right|$$

that there exists a C such that

$$K_\varphi(t) \leq \frac{C^p}{t^p}.$$

Nick's Question 1.9 Representation (1.19) from the wave paper I have a question about (1.19), here are my steps trying to prove it but I am unsure why the last step is true. For notation, let $\xi = (\xi_1, \dots, \xi_n)$ and $\mu(d\xi) = \mu(d\xi_1) \cdots \mu(d\xi_n)$. Also $x = (x_1, \dots, x_n)$

$$\begin{aligned} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \int_{\mathbb{R}^{nd}} \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi_1, \dots, \xi_n) \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi_1, \dots, \xi_n) \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &= \int_{\mathbb{R}^{nd}} \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi) \mathcal{F}(\tilde{f}_n(\cdot, 0, t))(\xi) \mu(d\xi) \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) \Gamma(dx) \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) \gamma(x) dx \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} [\tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t)](x) (K * K)(x) dx \\ &= (2\pi)^d \langle \tilde{f}_n(\cdot, 0, t) * \tilde{f}_n(\cdot, 0, t), K * K \rangle \\ &= (2\pi)^d \langle \tilde{f}_n(\cdot, 0, t) * K, \tilde{f}_n(\cdot, 0, t) * K \rangle \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left([\tilde{f}_n(\cdot, 0, t) * K](x) \right)^2 dx \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left(\int_{\mathbb{R}^{dn}} \tilde{f}_n(y, 0, t) K(x - y) dy \right)^2 dx \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left(\int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} f_n(y_{\sigma(1)}, \dots, y_{\sigma(n)}) K(x - y) dy \right)^2 dx \\ &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left(\int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} G(s_1, y_{\sigma(1)}) \cdots G(s_n - s_{n-1}, y_{\sigma(n)} - y_{\sigma(n-1)}) ds K(x - y) dy \right)^2 dx \end{aligned}$$

and therefore we have

$$\begin{aligned} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &= (2\pi)^d \int_{\mathbb{R}^{dn}} \left(\int_{\mathbb{R}^{dn}} \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} G(s_1, y_{\sigma(1)}) \cdots G(s_n - s_{n-1}, y_{\sigma(n)} - y_{\sigma(n-1)}) ds K(x - y) dy \right)^2 dx \\ &\quad (1.14) \quad \boxed{\text{rep}} \end{aligned}$$

This is pretty much (1.19) but if you look at (1.19) you can see that

$$K(x - y) = K(x_1 - y_1) \cdots K(x_n - y_n)$$

but why does this have to be true?

Nick's Question 1.10 Missing $(2\pi)^{-d}$ constant and definition of Fourier transform There are a couple places where this constant may be missing. This is because when applying Plancherel's Theorem with

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{-ix\xi} dx$$

we have

$$\langle f, g \rangle = (2\pi)^{-d} \langle \mathcal{F}(f), \mathcal{F}(g) \rangle$$

and this $(2\pi)^{-d}$ was ignored in this paper. If you see in 1.14, there is a $(2\pi)^d$ in the front. I also believe that there is some unnecessary work dealing with the $\mathcal{M}$ representation of ρ .

First, in the Xia Chen paper, we have

$$\Lambda_\alpha = \sup_{f \in \mathcal{F}_2} \left\{ \langle f^2 * f^2, \gamma \rangle^{1/2} + \mathcal{E}_2(f, f) \right\}$$

where

$$\mathcal{E}_2(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(x) dx$$

and Xia Chen proves that

$$\rho = (2\pi)^{-d} \Lambda_\alpha^{(4-\alpha)/2} \quad (1.15) \quad \text{xia rep}$$

Now, in the wave paper, we introduce the $\mathcal{M}$ functional

$$\mathcal{M}(f, \theta) = \sup_{g \in \mathcal{F}_2} \left\{ \langle g^2 * g^2, f \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx \right\}$$

because

$$\int_{\mathbb{R}^d} |\nabla g(x)|^2 dx = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 |\mathcal{F}g(x)|^2 dx$$

Why don't we just use 1.15? Or even just leave everything in terms of ρ ?

However, in the Xia Chen Paper he uses

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{ix\xi} dx$$

as the definition of the Fourier transform and so 1.15 is proven for that definition. In the wave paper,

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{-ix\xi} dx$$

is taken to be the definition. This means that 1.15 would need to be changed if we use this definition of the Fourier transform. Specifically, I think that all that needs to change is that we need to define

$$\mathcal{E}_2(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(-x) dx.$$

But by a quick change of variables its easy to see that

$$\int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(-x) dx = \int_{\mathbb{R}^d} |x|^2 \mathcal{F}(f)(x) dx$$

However, I think it would be more consistent if we follow Xia Chen and use his definition of the Fourier Transform.

Nick's Question 1.11 (6.9) verification that

For our definition of the Fourier we have

1. $\widehat{f * g} = \hat{f} \hat{g}$
2. $\widehat{fg} = (2\pi)^{-d} \hat{f} \hat{g}$
3. $\langle f, g \rangle = (2\pi)^{-d} \langle \hat{f}, \hat{g} \rangle$ Verification of (6.2) for $d = 2$. First we note that we are only now considering the Reisz Kernal, γ , which is an even function. Since G is also even then $G * \gamma$ is even which means $\mathcal{F}(G * \gamma)$ is even. Also we are only considering f to be non-negative and non-negative definite which implies that $\mathcal{F}f$ will be even as well.

$$\begin{aligned}
W_2(t, \mathcal{F}(f)) &= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \int_{\mathbb{R}^d} \mu(d\xi_1) \mathcal{F}f(\xi_1) \mathcal{F}f(\xi_2) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \int_{\mathbb{R}^d} \mu(d\xi_1) \mathcal{F}f(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \int_{\mathbb{R}^d} d\xi_1 \mathcal{F}(f * \gamma)(\xi_1) \mathcal{F}G(s_1, \cdot)(-\xi_1 + \xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} \mu(d\xi_2) \mathcal{F}f(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) [\mathcal{F}(f * \gamma) * \mathcal{F}G(s_1, \cdot)](\xi_2) \\
&= (2\pi)^{-d} \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} d\xi_2 \mathcal{F}(f * \gamma)(\xi_2) \mathcal{F}(G(s_2 - s_1, \cdot))(\xi_2) \mathcal{F}[(f * \gamma)G(s_1, \cdot)](\xi_2) \\
&= (2\pi)^{-d} \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} d\xi_2 \mathcal{F}[(f * \gamma) * G(s_2 - s_1, \cdot)](\xi_2) \mathcal{F}[(f * \gamma)G(s_1, \cdot)](\xi_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} dx_2 [(f * \gamma) * G(s_2 - s_1, \cdot)](x_2) [(f * \gamma)G(s_1, \cdot)](x_2) \\
&= \int_{[0,t]^2 <} ds \int_{\mathbb{R}^d} dx_2 (f * \gamma)(x_2) G(s_1, x_2) \int_{\mathbb{R}^d} dx_1 (f * \gamma)(x_1) G(s_2 - s_1, x_2 - x_1) \\
&= \int_{[0,t]^2 <} ds \iint_{\mathbb{R}^{2d}} dx_2 dx_1 (f * \gamma)(x_1) G(s_2 - s_1, x_2 - x_1) (f * \gamma)(x_2) G(s_1, x_2) \\
&= \int_{[0,t]^2 <} ds \iint_{\mathbb{R}^{2d}} dx_2 dx_1 (f * \gamma)(x_2) G(s_2 - s_1, x_2 - x_1) (f * \gamma)(x_1) G(s_1, x_1)
\end{aligned}$$

and the last line follows from a substitution $x'_1 = x_2$ and $x'_2 = x_1$ and the fact that $G(t, \cdot)$ is even.

2 Proof Sketches

Nick's Question 2.1 A mild solution for

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases}$$

must satisfy the following integral equation

$$u(t, x) = 1 + \int_0^t \left(\int_{\mathbb{R}^d} u(t-s, x-y) Y(s, y) W(dy) \right) ds$$

an through a Piccard Iteration Scheme we can see that the solution should be given by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.1) \quad \text{sol}$$

where

$$\begin{aligned} f_k(x_1, \dots, x_k, x, t) &= \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t-t_k, x-x_k) \cdots Y(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t_1, x-x_k) \cdots Y(t_k-t_{k-1}, x_2-x_1) dt_1 \cdots dt_n \end{aligned} \quad (2.2) \quad \text{solker}$$

$$(2.3)$$

and also

$$\mathcal{F}f_k(\cdot, x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]^{n+1}} \prod_{k=1}^n \overline{\mathcal{F}Y(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt \quad (2.4) \quad \text{fsolker}$$

where $t_{n+1} = t$

Nick's Question 2.2 Greens Function Scaling Property For and $c > 0$ we have

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi). \quad (2.5) \quad \text{fscale}$$

which is shown above.

Nick's Question 2.3 $\mathcal{H}^{\otimes n}$ Norm Scaling Property Suppose that $\mu(d\xi) = \varphi(\xi)d\xi$ and φ satisfies $\varphi(cx) = c^{-(d-\alpha)}$ for some $\alpha \in (0, d)$ and for any $c > 0$. Also recall that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi).$$

Then

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \quad (2.6) \quad \text{nscale}$$

Proof:

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, t)^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi}$$

where $\varphi(\vec{\xi}) = \varphi(\xi_1) \cdots \varphi(\xi_n)$

$$\begin{aligned}
& \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, t)^2 \varphi(\vec{\xi}) d\vec{\xi} = \int_{\mathbb{R}^{nd}} \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n <} \overline{\mathcal{F}Y(t - t_n, \cdot) \left(\sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)} \int_{\mathbb{R}^{nd}} \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n <} \mathcal{F}Y \left(1 - \frac{t_n}{t}, \cdot \right) (t^{b/a} \sum_{i=1}^n \xi_i) \cdots \mathcal{F}Y \left(\frac{t_2}{t} - \frac{t_1}{t}, \cdot \right) (t^{b/a} \xi_1) \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}Y(1 - t_n, \cdot) (t^{b/a} \sum_{i=1}^n \xi_i) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (t^{b/a} \xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}Y(1 - t_n, \cdot) \left(\sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(t^{-b/a} \vec{\xi}) t^{-ndb/a} d\vec{\xi} \\
& = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \left(\frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}Y(1 - t_n, \cdot) \left(\sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \mathcal{F} \tilde{f}_n(\cdot, 0, 1) (\vec{\xi})^2 \varphi(\vec{\xi}) d\vec{\xi} \\
& = t^{[2(b+r-1)+2-nb\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\
& = t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2
\end{aligned}$$

Nick's Question 2.4 Lemma 2.3 We define

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

and then we also define

$$\mathcal{M}_a(f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \right\} \quad (2.7) \quad \text{var}$$

Now we define $\tilde{g}(x) = B^{-d/2} g(x/B)$. Its easy to show that $\mathcal{F}\tilde{g}(\xi) = B^{d/2} \mathcal{F}g(B\xi)$. By a change of variables, we can then show that

$$\int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi = B^{-a} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi$$

Next, by applying the scaling property of f , $f(cx) = c^{-\alpha} f(x)$, we can show by again using a change of variables that

$$\begin{aligned}
(\tilde{g}^2 * f)(x) &= B^{-d} \int_{\mathbb{R}^d} \tilde{g}^2(y/B) f(x - y) dy \\
&= \int_{\mathbb{R}^d} g^2(u) f(x - uB) du \\
&= B^{-\alpha} \int_{\mathbb{R}^d} g^2(u) f\left(\frac{x}{B} - u\right) du \\
&= B^{-\alpha} (g^2 * f)(x/B)
\end{aligned}$$

This can be used to show that

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$$

Now we can rewrite (2.7) as the following

$$\mathcal{M}_a(\Theta f, \theta) = \frac{1}{\theta} \sup_{g \in \mathcal{F}_a} \left\{ \frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle^{1/2} - \frac{B^a}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi \right\}$$

and as before, in order to preserve a scaling property, we want to choose B such that

$$\frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} = B^a$$

and solving for B we get

$$B = \left(\frac{\sqrt{\Theta}}{\theta} \right)^{\frac{2}{2a-\alpha}}$$

and after plugging this B in we get that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1)$$

Some definitions for solvability (details in my notebook)

$$\begin{aligned} L_n(y_1, \dots, y_n) &= n! \int_0^\infty e^{-t} \tilde{f}_n(y_1, \dots, y_n, 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n <} \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) ds dt \end{aligned}$$

where $s_0 = y_{\sigma(0)} = 0$.

$$\begin{aligned} H_n(t, \mathbf{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \tilde{f}_n(y_1, \dots, y_n, 0, t) d\mathbf{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n <} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\mathbf{y} ds \end{aligned}$$

$$L(x) = \int_0^\infty e^{-t} Y(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}Y(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}$$

$$L_n(y_1, \dots, y_n) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)})$$

$$\mathcal{F}L_n(\xi_1, \dots, \xi_n) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}$$

We the two representations for T_n

$$T_n = \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) = \left(\frac{\nu}{2} \right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi})$$

We have the two possible representations for C'_μ .

$$C'_\mu = \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) = \left(\frac{\nu}{2} \right)^{-\alpha/a} \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi)$$

Nick's Question 2.5 Lemma 3.3, 3.4, and 3.5 Same proof as in wave paper. Have all details in my notebook.

Nick's Question 2.6 Lemma 3.6 We have that

$$C'_\mu < \infty \quad \text{and} \quad T_n < (n!)^2 (C'_\mu)^n$$

proof: By the same spherical substitution in the wave paper we see that

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = \left(\int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left(\int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho \right)$$

The first integral is shown to be finite in the wave paper. The second integral is finite if

$$2a - \alpha + 1 > 1 \tag{2.8}$$

alpha2

The same arguments from the wave paper show that

$$\begin{aligned} T_n &\leq n! \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &= (n!)^2 \int_{(\mathbb{R}^d)^n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &\leq (n!)^2 \left[\sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi + \eta|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 (C'_\mu)^n \end{aligned}$$

Nick's Question 2.7 Lemma 3.7

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} }{(n!)^2} T_n \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n$$

Proof: We first derive a scaling property for $\mathcal{F}\tilde{f}_n$ where $\tilde{f}_n$ is the kernel of the n^{th} Wiener integral in the chaos expansion of the solution. Let $c > 0$.

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$$

so it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually.

$$\begin{aligned}\mathcal{F}f(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(t_{k+1} - t_k, \cdot) \left(\sum_{j=1}^k \xi_j \right)} d\vec{t} \\ &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(c(t_{k+1} - t_k), \cdot) \left(\sum_{j=1}^k \xi_j \right)} c^n d\vec{t}\end{aligned}$$

where we applied the substitution $t'_i = t_i/c$. Now by applying the scaling property of $\mathcal{F}Y$ we see that

$$\begin{aligned}\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}Y(c(t_{k+1} - t_k), \cdot) \left(\sum_{j=1}^k \xi_j \right)} c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}Y(t_{k+1} - t_k, \cdot) \left(c^{b/a} \sum_{j=1}^k \xi_j \right)} c^n d\vec{t} \\ &= c^{n(b+r)} \mathcal{F}f(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n)\end{aligned}$$

Now by using this scaling property we see that

$$\begin{aligned}\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(2^{b/a} \xi_1, \dots, 2^{b/a} \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 2^{\frac{-nbd}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\ &= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \tilde{f}_n(\cdot, 0, t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt\end{aligned}$$

and now we apply Lemma 3.5 to get

$$\begin{aligned}\frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} \left(\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right)^2 d\vec{x} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\vec{\xi})|^2 d\vec{\xi} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n\end{aligned}$$

and this proves the first inequality. An application of Lemma 3.6 gives is the second inequality.

Nick's Question 2.8 Theorem 3.1 1.6 has a unique solution $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$ and

$$\sup_{(t, x) \in [0, T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \quad \forall p \geq 2 \text{ and } T > 0.$$

Proof: First using 2.5 (details in notebook) we can show that

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\tilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n). \quad (2.9)$$

In order to prove the theorem we need to show that $\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. From the scaling property 2.6 we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r) - \frac{b\alpha}{a}]n} dt \\ &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0 \quad (2.10) \quad \text{alpha3}$$

By using Lemma 3.7 we can deduce that

$$\begin{aligned} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \end{aligned}$$

By the same reasoning as in the proof in the wave paper (details in my notebook) we can say that there exists a constant depending on α , k_α , such that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}}$$

Therefore,

$$\begin{aligned} \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &\leq \sum_{n \geq 0} \theta^n n! k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}} \\ &= \sum_{n \geq 0} \theta^n k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}} \end{aligned}$$

and the last sum is finite when

$$2(b+r) - 1 - b\alpha/a > 0 \quad (2.11) \quad \text{alpha4}$$

Some definitions for the following lemma which come from the Xia Chen paper.

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right] dx$$

$$\mathcal{E}_a(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |x|^a |\mathcal{F}(x)|^2 dx$$

$$\begin{aligned}\Lambda_\alpha &= \sup_{f \in \mathcal{F}_a} \left\{ \left(\int_{\mathbb{R}^{2d}} \frac{f^2(x)f^2(y)}{|x+y|^\alpha} dx dy \right)^{1/2} - \mathcal{E}_a(f, f) \right\} \\ &= \sup_{f \in \mathcal{F}_a} \left\{ \langle f^2 * f^2, |\cdot|^{-\alpha} \rangle^{1/2} - \mathcal{E}_a(f, f) \right\}\end{aligned}$$

$$\rho = (2\pi)^{-d} \Lambda_\alpha^{(2a-\alpha)/a}$$

Nick's Question 2.9 Lemma 4.3

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} 2^{2(b+r) - \frac{b\alpha}{a}} \right) + \log(\rho) \quad (2.12) \quad \boxed{2ineqa}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{\nu}{2} \right)^{-\alpha/a} \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right] + \log(\rho) \quad (2.13) \quad \boxed{2ineqb}$$

proof: Denote T'_n to be T_n with $\nu = 2$, in other words

$$T'_n = \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n).$$

Recall Theorem 2.3 from the Xia Chen paper which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1}{(n!)^2} T'_n \right) \rightarrow \log(\rho)$$

Note that through a change of variables, $\xi_i = \left(\frac{\nu}{2} \right)^{1/a} \xi_i$, we can rewrite $T_n = \left(\frac{\nu}{2} \right)^{-n\alpha/a} T'_n$. We already showed in the proof of Theorem 3.1 above that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then Lemma 3.7 above shows that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]}}{(n!)^2} T_n = \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]}}{(n!)^2} \left(\frac{\nu}{2} \right)^{-n\alpha/a} T'_n.$$

Combining this and Theorem 2.3 from the Xia Chen paper we see that

$$\begin{aligned}\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) \\ \leq \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} 2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T'_n}{(n!)^2} \right) \\ = \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} 2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \log(\rho)\end{aligned}$$

which proves the upper bound for 2.12.

Now define for $t > 0$ and $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall Lemma 3.4 which says

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{\mathbb{R}^{nd}} H_n(t, x)^2 dx = \frac{1}{(n!)^2} J_n(t, t).$$

Thus $J_n(t, t)$ satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r)-b\alpha/a]n} J_n(1, 1).$$

Let τ and $\tilde{\tau}$ be independent exponential random variables with mean 1. By lemma 3.3,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) = T_n \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r)-b\alpha/a]n/2} t'^{[2(b+r)-b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[\tau^{[2(b+r)-b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r)-b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \end{aligned}$$

where the last line follows from Lemma 3.4. Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \frac{\left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad (2.14) \quad \boxed{2lowbd}$$

Now, another application of Stirling's formula shows that

$$\Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r)-b\alpha/a]n+1) 2^{-[2(b+r)-b\alpha/a]n} C_n$$

where $C_n = 2^{-1}[2(b+r)-b\alpha/a]^{1/2}(2\pi n)^{1/2}$. We now infer from 2.14 that

$$c_\alpha \frac{2^{[2(b+r)-b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n \leq \Gamma([2(b+r)-b\alpha/a]n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where c_α is a constant depending on α . Since $\frac{1}{c}C_n \rightarrow 0$ as $n \rightarrow \infty$, we obtain the lower bound

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma([2(b+r) - b\alpha/a]n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) \\ \geq \log \left(\left(\frac{\nu}{2} \right)^{-\alpha/a} 2^{[2(b+r) - b\alpha/a]} \right) + \log(\rho) \end{aligned}$$

Thus 2.12 is proven. 2.13 is shown in the same way as in the wave paper.

Nick's Question 2.10 Theorem 4.1 (p=2 case)

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left(\frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r) - b\alpha - a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned} \quad (2.15)$$

Proof:

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}$$

2.13 above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left[\left(\frac{\nu}{2} \right)^{-\alpha/a} \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right] \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{\nu}{2} \right)^{-\alpha/a} \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left(t^{(2(b+r) - b\alpha/a)} \right)^n = A$$

By lemma A3 notice that

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R) t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\begin{aligned} \lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R)t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} \\ = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \end{aligned}$$

hence we see that

$$\beta = \frac{2(b+r) - b\alpha/a}{2(b+r) - (b\alpha/a) - 1} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1](\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}}$$

Finally, lemma A2 says that

$$\begin{aligned} \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left(\frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned}$$

Nick's Question 2.11 Theorem 5.1 (Upper Bound)

$$\begin{aligned} \limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ \leq \left(\frac{1}{2} \right) \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left(\frac{\nu}{2} \right)^{\frac{-\alpha}{2a(b+r)-b\alpha-a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right) \end{aligned} \quad (2.16)$$

where β is defined in Theorem 4.1.

proof: Same as in wave paper.

Function needed for lower bound

$$W_n(t, \phi) := \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds$$

Nick's Question 2.12 Lemma 6.2

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n)$$

proof: Same proof as in wave paper. details in notebook

Nick's Question 2.13 Proposition 6.3

For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$$

proof: Same as wave paper.

In order to give a fractional version of Lemma 6.4 and Proposition 6.5, we need to derive the fractional version of the time scale distribution given at the top of page 26 of the wave paper. We start with a the general form

$$f_t(x) = t^V f(t^W x)$$

Note that for $\xi \in \mathcal{C}^d$

$$\mathcal{F}f_t(\xi) = t^{V-dW} \mathcal{F}f(t^{-W}\xi)$$

We follow out the steps of Lemma 6.4 using the f_t defined above and then end up at a series of two linear equations with V and W being the unknowns which allows us to find W and V . First we calculate $\|f_\tau\|_{\mathcal{H}}^2$.

$$\begin{aligned} \|f_\tau\|_H^2 &= \tau^{2(V-dW)} \int_{\mathbb{R}^d} |\mathcal{F}f(\tau^{-W}\xi)|^2 \phi(\xi) d\xi \\ &= \tau^{2V-dW} \int_{\mathbb{R}^d} |\mathcal{F}f(u)|^2 \tau^{-W(d-\alpha)} \varphi(u) du \\ &= \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \end{aligned}$$

Lastly, we calculate $W_n(t, \mathcal{F}f_\tau)$

$$\begin{aligned} W_n(t, \mathcal{F}f_\tau) &= \tau^{n(V-dW)} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(\tau^{-W}\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds \\ &= \tau^{n(V-W(d-\alpha))} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(u_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\tau^W(u_k + \dots + u_n)) \varphi(\mathbf{u}) d\mathbf{u} ds \\ &= \tau^{n(V-W(d-\alpha))} \tau^{-n\frac{a}{b}W(b+r-1)} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \mathcal{F}f(u_k) \prod_{k=1}^n \mathcal{F}Y(\tau^{\frac{aW}{b}}(s_k - s_{k-1}), \cdot)(u_k + \dots + u_n) \varphi(\mathbf{u}) d\mathbf{u} ds \\ &= \tau^{n(V-W(d-\alpha))} \tau^{-n\frac{a}{b}W(b+r-1)} \tau^{-n\frac{aW}{b}} W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f) \\ &= \tau^{n[V-W((d-\alpha)+\frac{a}{b}(b+r))]} W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f) \end{aligned}$$

In order to ensure Lemma 6.4 and Proposition 6.5 works in the fractional case we need to choose V and W so that

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) = 0 \\ 2(V - dW) + W\alpha = \beta \end{cases}$$

Doing this ensures that

$$\|f_t\|_{\mathcal{H}}^2 = t^\beta \|f\|_{\mathcal{H}}^2$$

and

$$W_n(t, \mathcal{F}f_\tau) = W_n(t\tau^{\beta-1}, \mathcal{F}f)$$

If we solve the above system of equations for V and W we see that

$$\begin{cases} V = (\frac{a}{b}(b+r) - \alpha + d) \frac{b}{a}(\beta-1) \\ W = \frac{b}{a}(\beta-1) \end{cases}$$

and with all of this we have the following fractional version of Lemma 6.4.

Nick's Question 2.14 Lemma 6.4 For

$$f_t(x) = t^{(\frac{a}{b}(b+r) - \alpha + d) \frac{b}{a}(\beta-1)} f\left(t^{\frac{b}{a}(\beta-1)} x\right)$$

we have that

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^\beta \|f\|_{\mathcal{H}}^2$$

and also

$$W_n(t, \mathcal{F}f_\tau) = W_n(t\tau^{\beta-1}, \mathcal{F}f)$$

Note that the above Lemma does reduce down to the wave equation. IE when $a = b = 2$ and $r = 0$ and $\beta = (4 - \alpha)/(3 - \alpha)$ then the above reduces down to f_t defined on the top of page 26 in the wave paper.

Nick's Question 2.15 Proposition 6.5 For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|$$

proof: We apply proposition 6.3 to f_τ with $\tau = (p - 1)t$ and use Lemma 6.4 which gives us

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} (q - 1)(p - 1)^\beta t^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t([p - 1]t)^{\beta-1}, \mathcal{F}f) \right|.$$

which finishes the proof since

$$(q - 1)(p - 1)^\beta t^\beta = (p - 1)^{\beta-1} t^\beta = t_p^\beta$$

Some preparation for Theorem 6.6. Recall that

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}Y(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds$$

and we now consider the space $\mathcal{H}_+ := \mathcal{H} \cap C_+(\mathbb{R}^d)$ where C_+ is the space of continuous non-negative and nonnegative definite functions on $\mathbb{R}^d$. For any $f \in \mathcal{H}_+$ we can show that

$$W_n(t, \mathcal{F}(f)) = \int_{[0, t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n Y(s_k - s_{k-1}, x_k - x_{k-1}) dx ds =: U_n(t, f)$$

Under assumptions on the parameters, the greens function for the fractional equation is non-negative hence under these conditions we get that $W_n(t, \mathcal{F}(f)) \geq 0$ for all the functions $f \in \mathcal{H}_+$.

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

Nick's Question 2.16 Rough sketch of Theorem 6.6 Mimicking the steps in the wave paper we start by letting τ be an exponential random variable with mean 1. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$ we have that $\mathcal{F}(f)$ is even and in addition, Lemma 6.2 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}(f)) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

I need to verify the details of the claim that by the Xia Chen paper that we can say that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n) = \log \rho(\mathcal{F}f)$$

where

$$\rho(g) = \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^a} \sqrt{1 + |\eta|^a}} d\eta \right] \mu(d\xi)$$

and with this being said, we see that for any $f \in \mathcal{H}_+$ and $\|f\|_H$

$$\liminf \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate $\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1}$. Consider a function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + |\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + |\cdot|^a}} \right) (x)$ is well defined. **I still need to verify the details that** under these conditions, $g \in W_{1,a}(\mathbb{R}^d)$ with $\|g\|_{W_{1,a}} = 1$. Here is a sketch of the details

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}f(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}f(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1 \end{aligned}$$

where above we used the fact that $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$.

Now, for notation let $\tilde{h}(x) = h(-x)$. Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^a} \sqrt{1 + |\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi - \eta) \mathcal{F}g(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \end{aligned}$$

Since $h(\cdot)$ is real value then $\widetilde{\mathcal{F}g} = \overline{\mathcal{F}g} = \mathcal{F}\bar{g}$. Using this we see that $(\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) = \mathcal{F}[|g|^2](\xi)$ and with this we see that

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} (f * \gamma)(x) |g|^2(x) dx \\ &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \langle f, |g|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}=1}} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}=1}} \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \langle f, |g|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal f is chosen to be $|g|^2 / \| |g|^2 \|_{\mathcal{H}}$. Therefore the supremum can be restricted to $\mathcal{H}_+$ which implies

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}=1}} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} (2\pi)^{-d} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2} &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |\mathcal{F}[|g|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g * \tilde{\mathcal{F}}g)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^a} \sqrt{1 + |\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \end{aligned}$$

which finally leads us to the inequality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2})$$

Nick's Question 2.17 Lemma 6.7 We have that

$$W_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} W_n(1)$$

and moreover

$$U_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} U_n(1)$$

proof: Let $\phi = \mathcal{F}f$ and $c > 0$

$$\begin{aligned} W_n(t, \phi(c^{-1} \cdot)) &= \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(c^{-1} \xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \varphi(\vec{\xi}) d\vec{\xi} d\vec{s} \\ &= c^{nd} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (c(\xi_k + \cdots \xi_n)) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y\left(c^{\frac{b}{a}} (s_k - s_{k-1}), \cdot\right) (\xi_k + \cdots \xi_n) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} c^{-n \frac{a}{b}} \int_{[0, c^{\frac{a}{b}} t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n Y(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \varphi(c\vec{u}) d\vec{u} d\vec{s} \\ &= c^{-n(a-\alpha+\frac{a}{b}r)} W_n\left(c^{\frac{a}{b}} t, \phi\right) \end{aligned}$$

Now we apply this with $c^{\frac{a}{b}} = \frac{1}{t}$ to get

$$W_n\left(t, \phi\left(t^{\frac{a}{b}} \cdot\right)\right) = t^{n \frac{b}{a} (a - \alpha + \frac{a}{b} r)} W_n(1, \phi)$$

Now note that

$$\left\| \phi(t^{\frac{a}{b}}) \right\|_{L^2(\mu)}^2 = t^{-\frac{\alpha b}{a}} \|\phi\|_{L^2(\mu)}^2$$

and from this we deduce that

$$\left\| t^{\frac{\alpha b}{2a}} \phi(t^{\frac{a}{b}}) \right\|_{L^2(\mu)}^2 = \|\phi\|_{L^2(\mu)}^2 = \|f\|_{\mathcal{H}}^2$$

and in addition

$$W_n \left(t, t^{\frac{b\alpha}{2a}} \phi(t^{\frac{a}{b}}) \right) = t^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} W_n(1, \phi)$$

and the result follows after taking the sup over all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} = 1$.

Nick's Question 2.18 Lemma 6.8

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right] \geq \log \left[\frac{\rho^{1/2}}{\left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right]$$

proof: Let τ be an exponential random variable with mean 1. By lemma 6.7,

$$\mathbb{E}(U_n(\tau)) = \mathbb{E} \left(\tau^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right) = \Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1)$$

and by Theorem 6.6 we obtain

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log(\rho^{1/2}). \quad (2.17) \quad \boxed{\text{lb1}}$$

Recall that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an+1)}{(n!)^a} \right) = a \log(a)$$

and with this we have that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right)}{(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] = \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log \left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)$$

and the result is proven after multiplying and dividing $(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}$ on the inside of the log in 2.17

Nick's Question 2.19 Lemma 6.9

For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = [c_1 t]$, it holds that

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \left(\rho^{1/2} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \quad (2.18) \quad \boxed{69ineq}$$

proof: Let $\epsilon > 0$. Lemma 6.8 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho^{1/2}}{\left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

Now fix a $c > 0$ and let $n_t := [ct]$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$.

Let $t > t_\epsilon$.

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{a}{b}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{a}{b}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} R^{n_t} e^{-n_t \epsilon}$$

Note that $[ct]/t \rightarrow c$ and $t \rightarrow \infty$ and this implies that $n_t/t \rightarrow c$ and $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t} \frac{n_t^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)n_t}}{(n_t!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

We can easily find the maximal c by solving $h'(c) = 0$ and doing so we get $c = (k\sqrt{\theta}R)^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$. After plugging in this value for c and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}} (\rho^{1/2})^{\frac{a}{b(a-\frac{\alpha}{2}+\frac{a}{b}r)}}$$

and simplifying exponents gives us [2.18](#).